Understanding Aadhaar Enrolment and Update Trends Across India

Data Driven Insights for Service Optimisation

UIDAI Data Hackathon 2026

Team ID: UIDAI_7876
Tools Used: Python, Pandas, Matplotlib, Jupyter Notebook

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1. Problem Statement and Approach

Problem Statement

Aadhaar is one of the world's largest digital identity systems, serving over a billion people across a diverse and mobile population. While enrollment numbers tell us how many people have joined the system, they don't capture the ongoing demand for services—like when someone updates their address after moving cities or refreshes their biometrics.

Understanding how enrollments and updates differ across age groups and regions helps us spot where the system faces the most pressure, where resources might be stretched, and how life events drive service needs. But raw numbers alone don't tell the full story, especially when update activity doesn't match enrollment levels.

This project aims to find meaningful patterns in anonymized Aadhaar data—enrollments, demographic updates, and biometric updates—to support smarter decision-making for service improvements and resource planning, without ever tracking individuals.

Approach

The analysis uses a structured, data-driven approach with aggregated and anonymized datasets from UIDAI. We focus on simple, clear techniques to extract insights that can actually inform policy decisions.

First, we look at enrollment activity across different age groups and regions to see which demographics are most active and where enrollments are concentrated. Then, we examine demographic and biometric update volumes to understand how administrative workload is distributed across states.

To meaningfully compare enrollments with updates, we created an "update intensity ratio"—basically measuring how often people update their information relative to enrollment numbers. This helps identify regions where people interact more frequently with Aadhaar services, possibly due to migration, life changes, or higher service demand.

We also explore how demographic and biometric updates relate to each other, revealing lifecycle-driven patterns across states. Throughout the analysis, we use clear visualizations to make insights easy to understand.

Our approach emphasizes transparency, reproducibility, and responsible data use, focusing on improving the system—not monitoring individuals.

2. Datasets Used

This analysis uses aggregated and anonymized Aadhaar datasets provided by UIDAI for the Data Hackathon 2026. The data contains no personal information and is designed for understanding population-level patterns only.

Aadhaar Enrolment Dataset

The enrollment dataset shows how many people enrolled in Aadhaar across different locations and age groups. Key information includes:

- When they enrolled
- Their state and district
- PIN code
- Age categories:
  - Age 0–5 years (young children)
  - Age 5–17 years (school-age kids and teens)
  - Age 18 years and above (adults)

We use this data to understand enrollment patterns across different age groups, see which regions have the most enrollments, and compare these baseline numbers with how often people update their information.

Aadhaar Demographic Update Dataset

The demographic update dataset tracks how often people update their Aadhaar information like addresses or phone numbers. Key details include:

- When the update happened
- State and district
- PIN code
- Age-wise update counts

This data helps us understand how updates are distributed across states and age groups, showing where administrative workload is highest due to people moving, changing jobs, or updating contact details as their lives change.

Aadhaar Biometric Update Dataset

The biometric update dataset tracks when people update their fingerprints or iris scans, either to refresh them or fix issues. Key information includes:

- When the update occurred

- State and district
- PIN code
- Age-wise update counts

We use this data to see where biometric updates happen most frequently and understand how they relate to demographic updates—like whether people tend to update both together.

Dataset Integration and Usage

We analyze these datasets both separately and together at the state level to understand patterns. Enrollment numbers serve as our baseline reference point, while demographic and biometric update volumes help us measure how intensively different regions use Aadhaar services and where administrative demand is highest.

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3. Methodology

The analysis follows a clear and transparent approach designed to find meaningful patterns in aggregated and anonymized Aadhaar data. We prioritize clarity, reproducibility, and responsible data use, relying on straightforward descriptive and comparative methods rather than complicated models.

Data Loading and Preprocessing

All datasets were imported using Python and processed using the Pandas library. Initial preprocessing steps included:

- Verifying dataset structure and column availability
- Converting date fields into appropriate date formats
- Checking for missing or inconsistent values
- Ensuring consistent geographic identifiers such as state names

As the datasets were pre-aggregated and anonymised, no individual-level cleaning or imputation was required.

Univariate Analysis

We started by examining each variable separately to understand its distribution. This included:

- Aadhaar enrollments across different age groups
- Total enrollments in each state
- Demographic updates by age group
- Biometric updates by age group

We used bar charts and time summaries to identify which age groups are most active, where enrollments are concentrated, and how different datasets compare in scale.

Bivariate Analysis

To explore how variables relate to each other, we conducted state-level comparisons. This included:

- Comparing total enrollments with demographic update volumes
- Comparing demographic and biometric updates across states

These comparisons helped us spot states where people update their information unusually often compared to enrollment numbers.

Update Intensity Ratio

To meaningfully compare enrollments with updates, we created an "update intensity ratio." This measures how many demographic updates happen relative to total enrollments in each state.

This indicator shows us which regions interact more frequently with Aadhaar services. High ratios suggest areas with more people moving around, experiencing life changes, or needing to update their information often.

Visualisation and Interpretation

We created visualizations using Matplotlib to clearly communicate what we found. Charts were designed to be simple and easy to understand, focusing on:

- Comparing relative differences rather than exact numbers
- Clear labels and consistent scales
- Patterns that matter for policy decisions rather than complex statistics

We interpreted findings at the system level, never looking at individuals.

Ethical Considerations

All analysis used aggregated and anonymized data only. No personal information was accessed or inferred. Our approach prioritizes responsible data use, focusing entirely on improving the system and planning services—never monitoring individuals.

4. Data Analysis and Visualisation

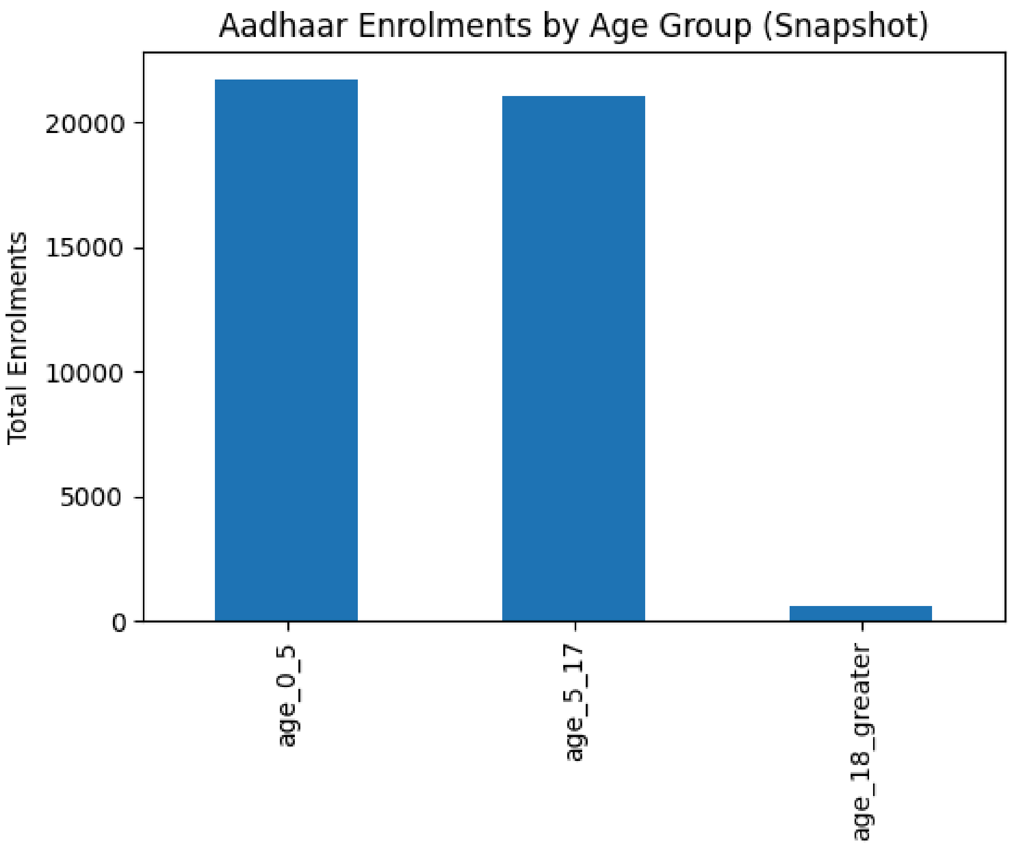
This section presents key findings from our analysis of Aadhaar enrollment, demographic update, and biometric update data. We use visualizations to highlight patterns and insights that can help with service planning.

Age-wise Aadhaar Enrollment Patterns

Looking at enrollment by age group, we found that most Aadhaar registrations happen among children aged 0–5 years and 5–17 years. Enrollments for people 18 and older are much lower in comparison.

This tells us that current enrollment activity is mainly driven by parents registering their young children—either as infants or during school years. Adult enrollments contribute very little to overall numbers, meaning enrollment infrastructure demand comes primarily from child and adolescent registrations.

Figure 1: Age-wise Aadhaar Enrolments Across India

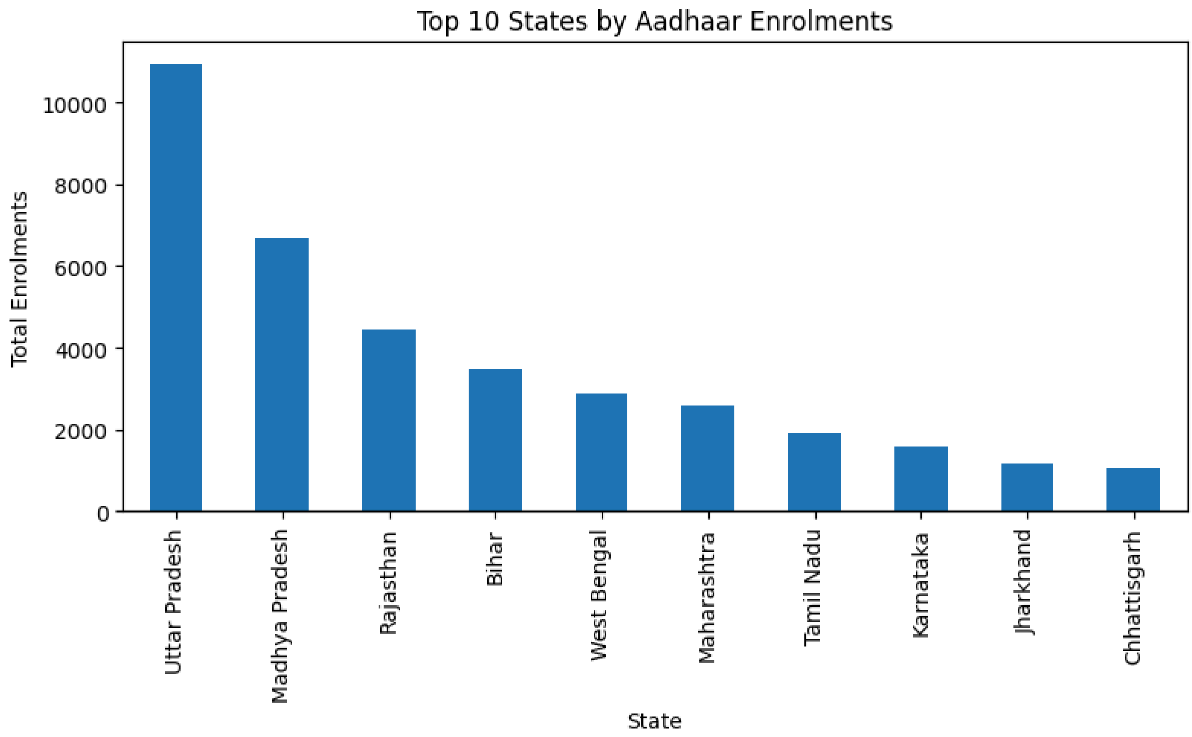


State-wise Distribution of Enrollments

When we look at enrollments by state, we see strong regional concentration. A handful of large states—like Uttar Pradesh, Madhya Pradesh, Rajasthan, Bihar, and Maharashtra—account for a huge portion of total Aadhaar enrollments.

This uneven distribution means larger states put much greater pressure on enrollment infrastructure and administrative resources.

Figure 2: Top States by Total Aadhaar Enrolments

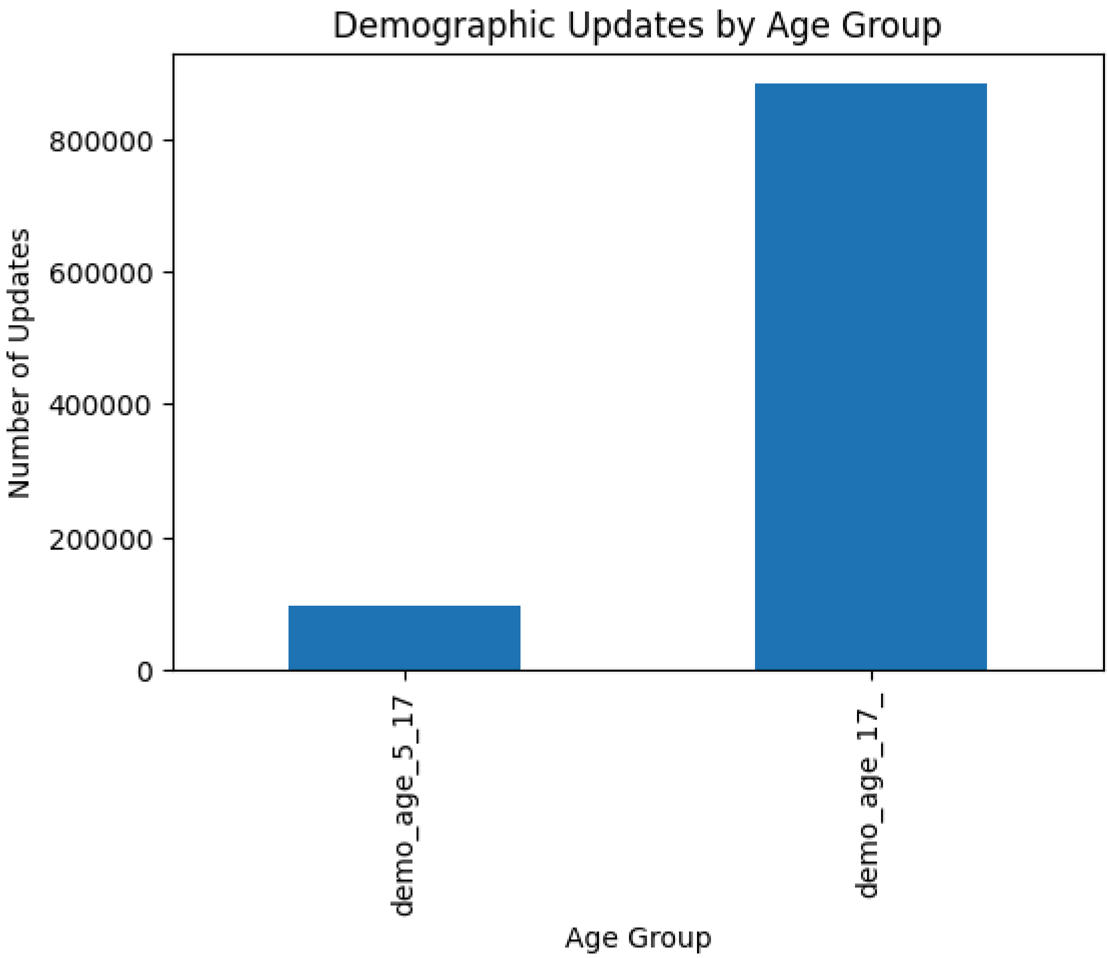


Demographic Update Activity Across Age Groups

Demographic updates (like changing your address or phone number) happen far more often among older age groups than younger ones. This makes sense—adults move cities for work, change jobs, get married, or experience other life events that require updating information.

The contrast between enrollment-heavy younger groups and update-heavy older groups shows why it's important to analyze updates separately from enrollments.

Figure 3: Demographic Updates by Age Group

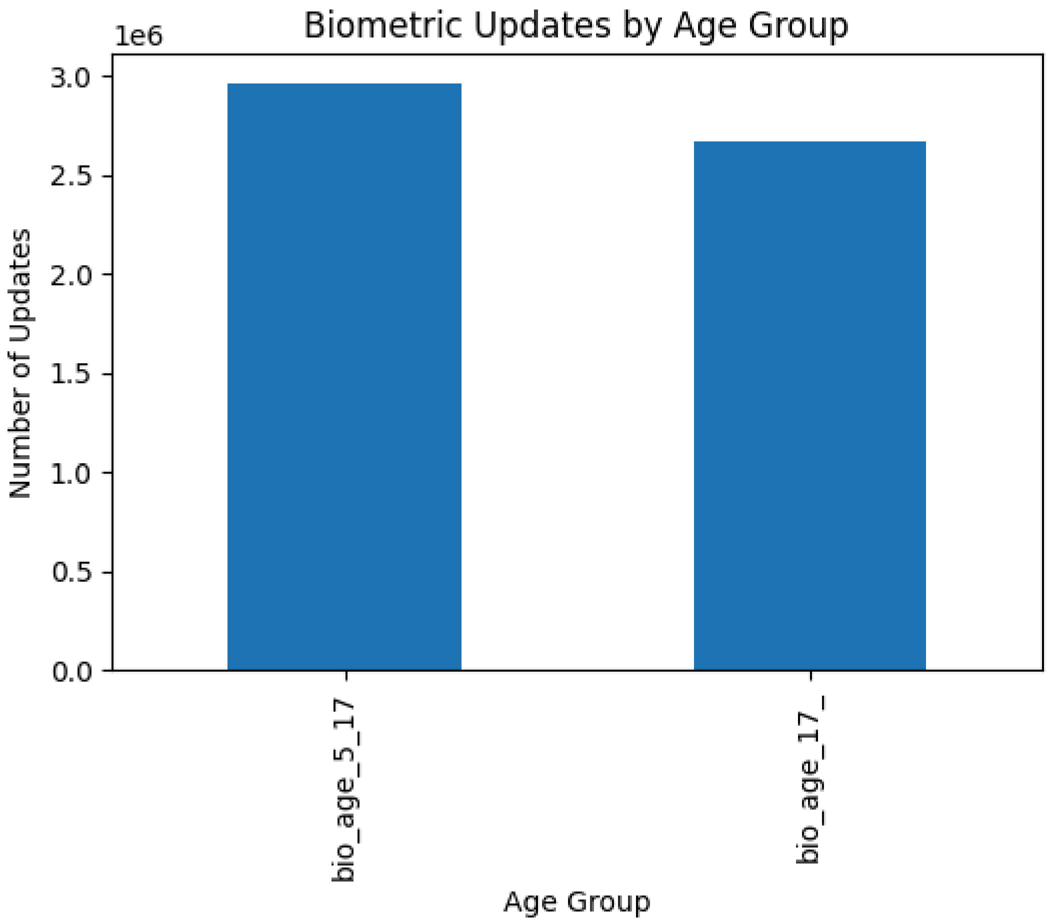


Biometric Update Patterns

Biometric updates (refreshing fingerprints or iris scans) also show significant activity across age groups, with patterns reflecting lifecycle-driven needs. Younger children may need updates as they grow and their features change, while adults might refresh biometrics due to fingerprint wear from manual work, aging effects, or to improve authentication accuracy.

The volume of biometric updates varies considerably by state, indicating different levels of administrative interaction and service use. States with higher activity might have populations more dependent on Aadhaar-linked services for banking or government programs. Regional differences could also reflect varying awareness about biometric refresh options or accessibility of update centers.

Figure 4: Biometric Updates by Age Group

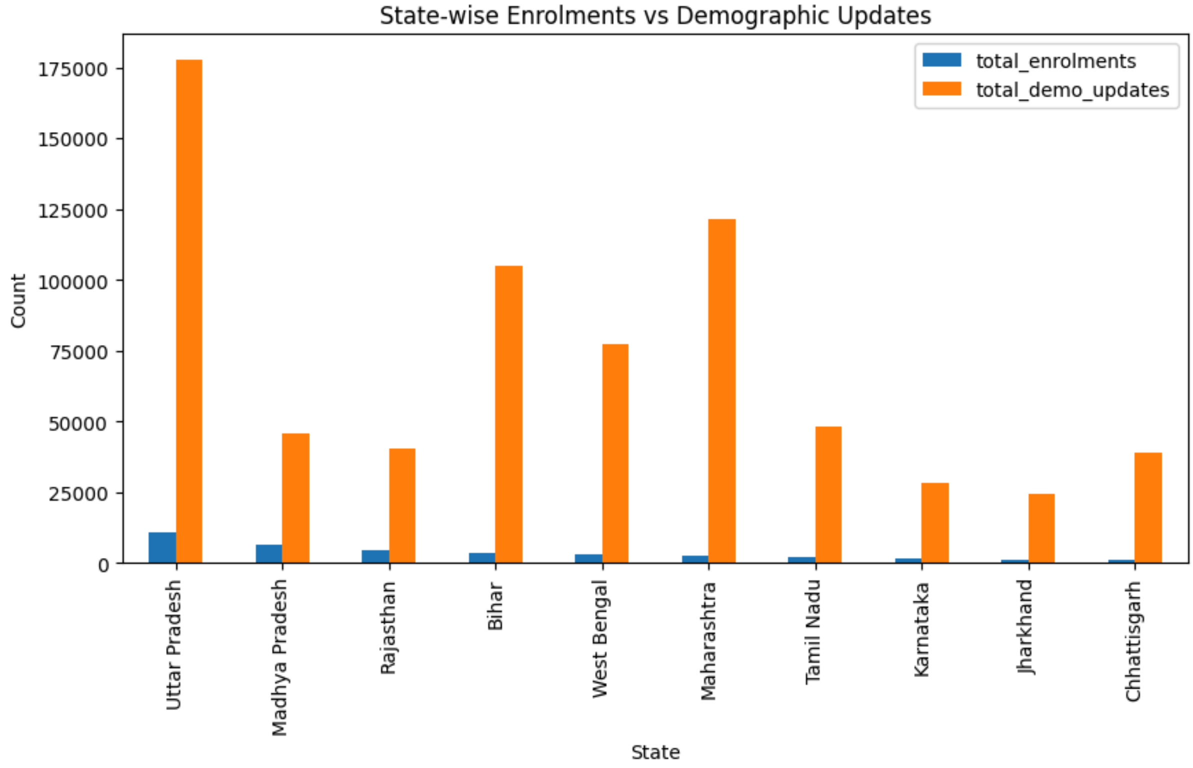


Enrollments Versus Demographic Updates

When we directly compare state-wise enrollments with demographic updates, something interesting emerges: update volumes don't increase proportionally with enrollment numbers. Some states with moderate enrollments show surprisingly high update activity, while others with large enrolled populations have relatively fewer updates.

This tells us that factors like population mobility, urbanization, and life transitions significantly influence service demand beyond just initial enrollment. For example, states serving as migration hubs for employment or education may see more people frequently updating addresses and contact details. Similarly, regions with younger, more mobile populations—students and workers relocating—naturally generate higher update demand. This suggests that infrastructure needs depend not just on enrollment size, but on how dynamic and mobile those populations are.

Figure 5: State-wise Comparison of Aadhaar Enrolments and Demographic Updates

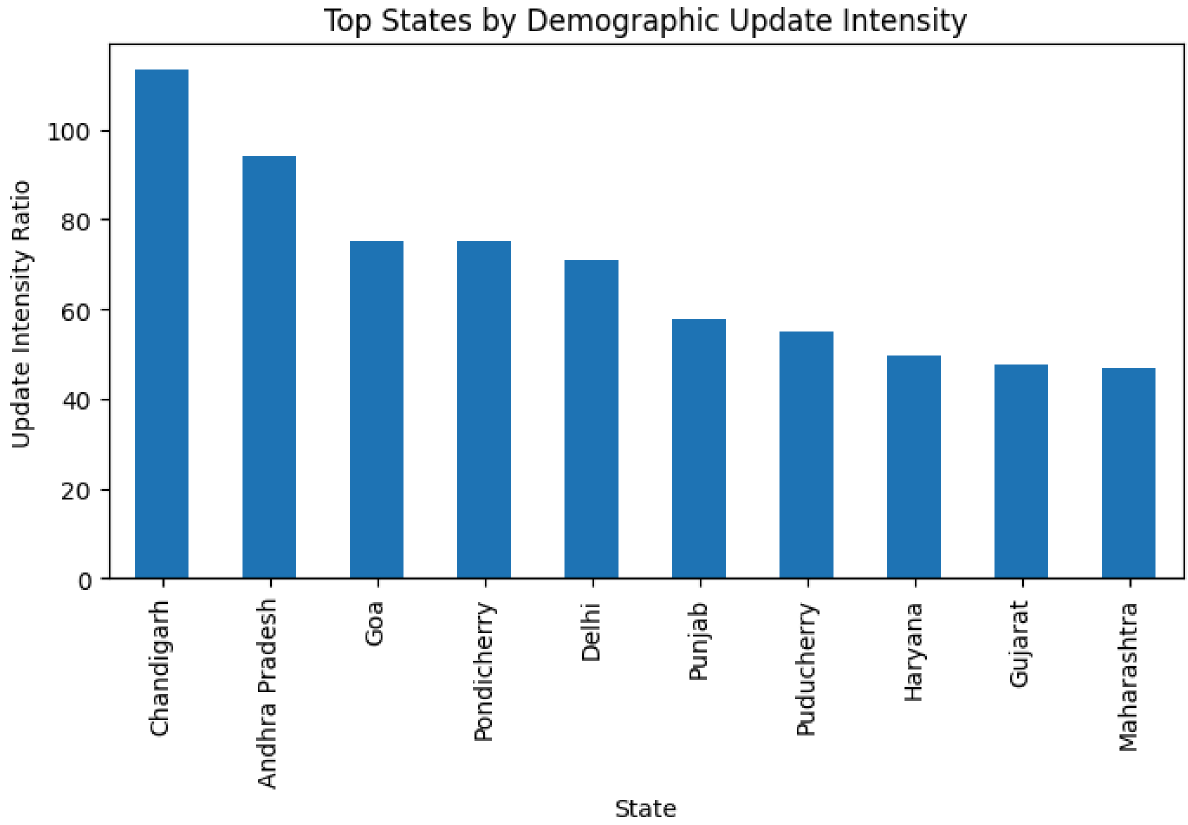


Update Intensity Ratio

To capture this pattern, we calculated an "update intensity ratio"—comparing demographic update volumes to total enrollments at the state level. States with higher ratios show more frequent interaction with Aadhaar services per enrolled person, indicating they're not just large in population but also more administratively active.

High update intensity might reflect several underlying factors: more migration as people move between cities for work, frequent address changes in urbanizing regions, career transitions requiring updated contact details, or greater reliance on Aadhaar-linked services like banking, subsidies, and government schemes that necessitate keeping information current. These states may face increased operational pressure—longer wait times at enrollment centers, higher staff workload, greater infrastructure demand—even if they don't have the highest absolute enrollment numbers. This makes the update intensity ratio a valuable indicator for identifying where UIDAI should prioritize resources, not just based on population size but on actual service interaction patterns.

Figure 6: Top States by Demographic Update Intensity Ratio



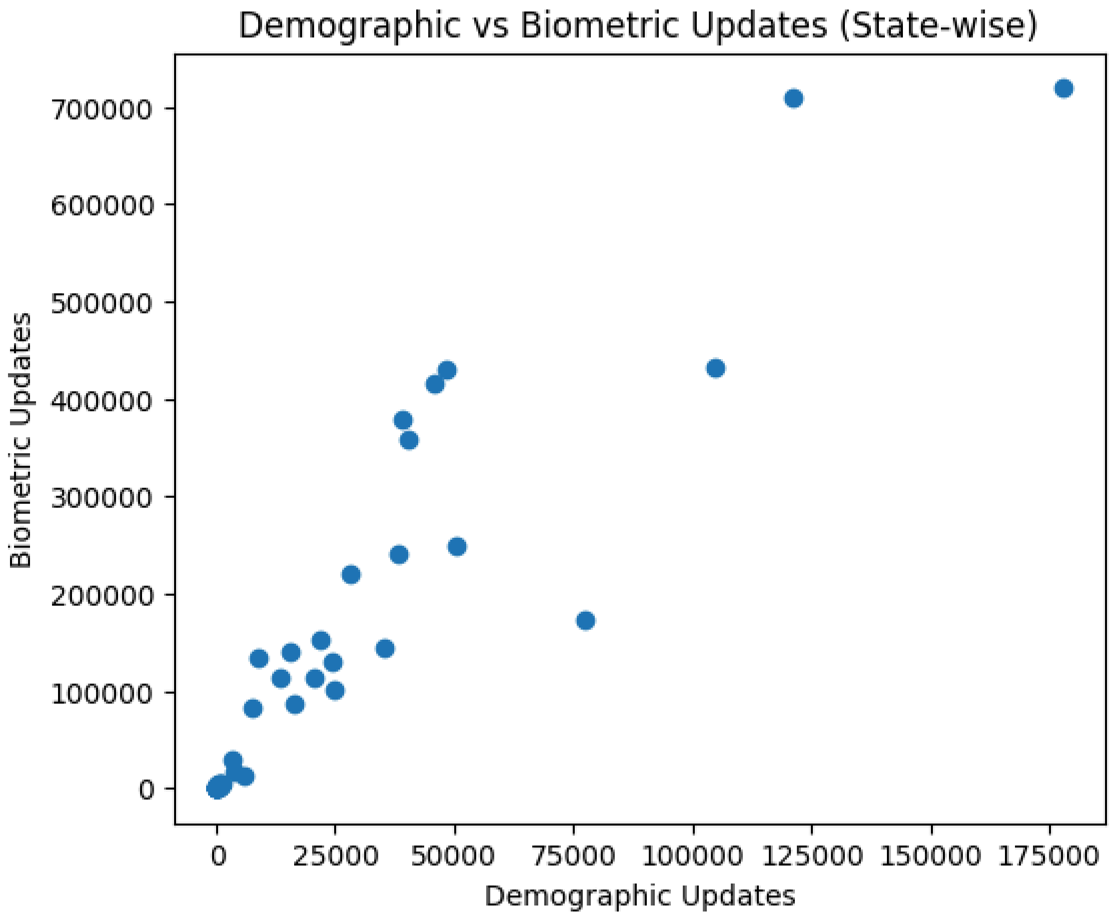
Relationship Between Demographic and Biometric Updates

We found a positive connection between demographic and biometric updates across states. Regions with more demographic update activity also tend to have more biometric updates, suggesting these two types of updates often happen together.

This correlation suggests lifecycle-driven behavior—when people's lives change enough to update their address or contact details, they often refresh their biometrics too. For instance, someone moving to a new city for work might update their address and simultaneously refresh fingerprints that have degraded. Major life events like marriage, job changes, or relocation often trigger comprehensive updates rather than isolated changes.

This pattern reinforces the need for integrated planning of update services. UIDAI centers should be equipped to handle both types of updates simultaneously, reducing multiple visits and improving user experience. States with high demographic update demand will likely need proportional biometric update infrastructure as well.

Figure 7: Demographic and Biometric Update Relationship by State



Summary of Insights

Overall, the analysis shows that Aadhaar service demand isn't just about enrollment volume—it's also shaped by update intensity driven by mobility and life changes. Visual analytics help reveal these patterns clearly, supporting data-driven identification of regions that may need enhanced operational capacity and targeted service improvements.

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5. Conclusion

This analysis used anonymized and aggregated Aadhaar enrollment, demographic update, and biometric update data to uncover meaningful patterns in how people across India use the Aadhaar system. Rather than just looking at enrollment numbers, we examined how update behavior varies by age group and region, revealing important insights about system usage and where administrative demand is highest.

The findings show that Aadhaar enrollment is largely driven by children and adolescents—parents registering their kids early. However, ongoing interaction with the system is strongly shaped by life events like moving cities for work or education, job changes, and other major transitions. Several states show high update volumes despite moderate enrollment levels, telling us that service demand isn't just about population size—it's also about how mobile and dynamic those populations are.

The "update intensity ratio" we introduced provides a simple but effective way to identify regions where Aadhaar services are used more frequently relative to enrollment volume. This helps distinguish between areas that mainly need enrollment infrastructure and those requiring sustained capacity for handling updates.

Administrative and Social Impact

The insights from this study can help UIDAI with:

- Identifying states facing higher workload due to frequent updates
- Optimizing placement of enrollment centers and update facilities
- Improving staffing and appointment planning in high-demand regions
- Reducing service bottlenecks and wait times
- Supporting smarter infrastructure planning at the state level

By focusing on aggregated patterns rather than individual behavior, the analysis supports responsible data use while improving service efficiency.

Limitations

This study uses aggregated and anonymized data, which means we can't make individual-level inferences. The analysis also represents a snapshot in time and doesn't capture long-term trends or seasonal variations. External factors like policy changes, regional events, or service availability weren't explicitly considered.

Future Scope

Future work could extend this framework by:

- Incorporating multi-year data to analyze trends over time
- Performing district- or PIN-code-level analysis for more detailed planning
- Integrating service infrastructure data like number of enrollment centers
- Developing interactive dashboards for real-time monitoring and decision support

Such extensions could transform this analytical approach into a predictive planning tool for large-scale identity systems like Aadhaar.

Ethical Note

All analysis in this project uses anonymized and aggregated data only. No personal information was accessed, inferred, or analyzed. The objective is system-level optimization and public service improvement—never individual monitoring.

Appendix A: Analysis Code Snippets

This appendix shows representative code snippets from our analysis, demonstrating how we loaded data, aggregated information, made comparisons, and created visualizations. The complete analysis was done using Python in Jupyter Notebook with Pandas and Matplotlib.

A.1 Data Loading

The anonymised Aadhaar datasets were loaded using Pandas from preprocessed CSV files. These datasets include enrolment data, demographic update data, and biometric update data.

```
import pandas as pd
import matplotlib.pyplot as plt

enrolment = pd.read_csv("../data/enrolment_cleaned.csv")
demographic = pd.read_csv("../data/demographic_cleaned.csv")
biometric = pd.read_csv("../data/biometric_cleaned.csv")
```

This step ensures structured access to all datasets for further aggregation and analysis.

A.2 State-wise Enrolment Aggregation

Enrolment data was aggregated at the state level by summing age-wise enrolment counts. A total enrollment column was derived to enable ranking and comparison across states.

```
state_enrol = enrolment.groupby('state')[['age_0_5','age_5_17','age_18_greater']].sum()
state_enrol['total'] = state_enrol.sum(axis=1)

top_states = state_enrol.sort_values('total', ascending=False).head(10)
```

This aggregation identifies states with the highest Aadhaar enrolment volumes.

A3. Enrolment Visualisation

A bar chart was generated to visualise the top states by total enrolments.

```
top_states['total'].plot(kind='bar', figsize=(9,4))
plt.title("Top 10 States by Aadhaar Enrolments")
plt.ylabel("Total Enrolments")
plt.xlabel("State")
plt.show()
```

The chart highlights regional concentration of enrolment demand.

A4. Age-wise Enrolment Distribution

To understand enrolment patterns by age group, enrolments were summed across all regions

```
age_totals = enrolment[['age_0_5','age_5_17','age_18_greater']].sum()

age_totals.plot(kind='bar', figsize=(6,4))
plt.title("Aadhaar Enrolments by Age Group (Snapshot)")
plt.ylabel("Total Enrolments")
plt.show()
```

This visualisation shows that enrolment activity is primarily driven by children and adolescents.

A5. Demographic and Biometric Update Comparison

Biometric update data was aggregated at the state level and compared with demographic updates to analyse lifecycle-driven system usage.

```
# Biometric updates per state
bio_cols = [col for col in biometric.columns if col.startswith('bio_')]
state_bio = biometric.groupby('state')[bio_cols].sum()
state_bio['total_bio_updates'] = state_bio.sum(axis=1)

# Combine with demographic updates
update_compare = state_demo[['total_demo_updates']].join(
    state_bio[['total_bio_updates']], how='inner'
)
```

A6. Update Relationship Visualisation

A scatter plot was used to visualise the relationship between demographic and biometric updates across states.

```
# Scatter plot
plt.figure(figsize=(6,5))
plt.scatter(
    update_compare['total_demo_updates'],
    update_compare['total_bio_updates']
)
plt.xlabel("Demographic Updates")
plt.ylabel("Biometric Updates")
plt.title("Demographic vs Biometric Updates (State-wise)")
plt.show()
```

This comparison reveals a positive association between demographic and biometric updates, suggesting lifecycle-linked update behaviour.

Note on Reproducibility and Ethics

All analysis was performed on aggregated and anonymised datasets. No individual-level data was accessed or inferred. The code focuses on transparency, reproducibility, and responsible use of public data.